



Design and Implementation of a Voice-Based Electronic Results System for Physically Challenged Students

Olayode Enoch Oladapo^{*1}, Ajayi Olusola Olajide², Adegbite Adewuyi Adetayo³, Akinwumi David Adeola⁴, Aju Omojokun Gabriel⁵, Akinrolabu Olatunde David⁶

¹Department of Computer Science, Faculty of Computing, Adekunle Ajasin University, Akungba-Akoko

²Department of Software Engineering, Faculty of Computing, Adekunle Ajasin University, Akungba-Akoko

^{3,6}Department of Data Science and Artificial Intelligence, Faculty of Computing, Adekunle Ajasin University, Akungba-Akoko

⁴Department of Cybersecurity, Faculty of Computing, Adekunle Ajasin University, Akungba-Akoko

⁵Department of Information Systems, Faculty of Computing, Adekunle Ajasin University, Akungba-Akoko

Corresponding Author: olayodeenoch@gmail.com

Article Info

Keywords: Voice-based system, speech recognition, artificial neural networks (ANN), hidden Markov model (HMM), result processing system, physically challenged students, inclusive education.

Received 10 March 2025

Revised 08 September 2025

Accepted 20 September 2025

Available online 17 November 2025



<https://doi.org/10.37933/nipes/7.4.2025.1324>

eISSN-2682-5821, pISSN-2734-2352

© 2025 NIPES Pub. All rights reserved.

Abstract

The growing population of physically challenged students in academic institutions necessitates the development of inclusive digital solutions. This research focuses on designing and implementing a voice-based electronic results system for Adekunle Ajasin University's existing result processing system, AVERS (Ajasin Varsity Examination Result System). The proposed system leverages Artificial Neural Networks (ANN) for speech recognition and automation and integrates Hidden Markov Models (HMMs) for sequential speech processing to provide hands-free operations for students, particularly those with dyslexia and other physical impairments. Unlike conventional result processing systems, this solution uniquely combines ANN and HMM in a hybrid architecture tailored specifically for academic environments, ensuring accurate speech recognition even in noisy university settings and accommodating diverse speech patterns among users. This adaptive approach enhances system robustness and user-friendliness, making it distinct from general-purpose voice systems such as Google Voice. This paper presents the problem, system design, and methodology, culminating in a comprehensive evaluation of the voice-enabled solution. The findings, which show a commendable accuracy rate of 92% in recognizing voice commands and an average response time of 2 seconds, demonstrate that incorporating voice automation significantly enhances the inclusiveness and usability of the institution's result-processing system, fostering independence, privacy, and improved accessibility for physically challenged students.

1.0. Introduction

The World Report on Disability [1], jointly presented by the World Health Organization (WHO) and the World Bank, reveals that 15% of the global population, approximately 1 billion people, experience some form of disability. This number continues to rise due to various factors such as road accidents, aging populations, and disease outbreaks, which often lead to physical impairments. Among this demographic, individuals with physical disabilities frequently face challenges in accessing essential services independently, including academic resources.

In educational institutions, students with physical disabilities often encounter difficulties in accessing their academic records without third-party assistance. Adekunle Ajasin University, Akungba Akoko, Ondo State, Nigeria, addresses result processing through the Ajasin Varsity Examination Result System (AVERS), a web-based solution developed using the Laravel PHP framework. AVERS allows staff to upload and manage student results while enabling students to securely access their academic reports. Despite its effectiveness, AVERS currently lacks

accessibility features tailored for students with disabilities, thereby limiting their independence in accessing academic records.

Speech recognition technology that enables hands-free interactions with digital systems presents a viable solution. For instance, AI-driven voice assistants have been successfully integrated into assistive technologies for visually impaired individuals, improving their engagement with digital platforms. This study focuses on designing and implementing a voice-based result system that integrates seamlessly with AVERS, providing physically challenged students with the ability to access their academic results independently.

The adoption of AVERS at Adekunle Ajasin University has significantly improved result processing efficiency and privacy for students. However, physically challenged students, particularly those with visual impairments, motor disabilities, or learning disorders such as dyslexia, face significant barriers in independently accessing their academic records. This reliance on third-party assistance compromises the privacy and autonomy that AVERS seeks to provide.

The absence of inclusive features in AVERS underscores the need for a voice-based solution that allows physically challenged students to access their results independently. By integrating voice automation, this research aims to enhance accessibility and inclusivity within the university's academic system. This research aims to enhance the existing AVERS system by integrating voice-based automation, making it more accessible to physically challenged students. The primary objectives include improving AVERS to accommodate their specific needs, incorporating voice automation for all system operations to promote inclusiveness, and developing a robust noise-filtering mechanism to enhance voice command accuracy.

The proposed system offers several key benefits. First, it enables physically challenged students to access their academic results independently, eliminating the need for third-party assistance. Additionally, the integration of voice automation will enhance overall usability, making AVERS more user-friendly for all students. Ultimately, this advancement fosters a more inclusive academic environment by addressing accessibility challenges and ensuring equal opportunities for all learners. Unlike conventional result processing systems, this study integrates an adaptive ANN-based speech recognition module tailored for academic environments, addressing both accessibility challenges and real-world usability constraints.

2. Literature Review

2.1 Electronic Result Processing Systems

Electronic result processing systems have become essential in managing academic records efficiently and securely. Amadin and Ukaoha [2] developed a computerized result system for the Computer Science Department at the University of Benin, emphasizing the use of object-oriented software design to streamline result management. Their findings highlighted significant improvements in administrative efficiency and data security but revealed a lack of features catering for physically challenged students. Bharamagoudar et al. [3] proposed a simple interface for maintaining students' information, which emphasized data integrity and minimized human errors associated with manual result processing. Similarly, Ekpenyong [4] developed an Intelligent Knowledge-Based System (IKBS) for result processing using Microsoft Excel and programming tools to improve data consistency. These systems lacked inclusivity features for physically challenged students. The introduction of automated result systems has transformed educational institutions by reducing the administrative burden and enhancing student privacy. However, these systems often overlook the needs of physically challenged students, necessitating more inclusive digital solutions.

2.2 Voice-Based Systems

Voice-based systems have gained prominence as an interface for hands-free interaction with digital devices. Aditi et al. [5] highlighted the integration of voice recognition with home automation

systems, making them more user-friendly for physically challenged individuals. Their study demonstrated that voice recognition significantly reduces operational barriers in everyday tasks. While existing research on voice-based educational systems [6], [5] has demonstrated potential benefits, studies rarely address the challenges of noisy environments in university settings. Moreover, past research [7] primarily focused on physical mobility applications rather than integrating voice recognition into academic management systems. This paper bridges that gap by developing an ANN-enhanced speech recognition system tailored for academic result retrieval.

2.3 Related Works

2.3.1 Academic Result Systems for Physically Challenged Students

Onyesolu and Ezenwegbu [8] proposed a computer-based testing system with voice commands designed for examination environments in Nigeria. Their findings indicated that incorporating voice commands into academic systems could improve accessibility for physically challenged students. However, their research was limited to test-taking scenarios and did not extend to result management systems. Ekpenyong [4] developed a result computation system for tertiary institutions but highlighted its limitations in accommodating visually impaired and motor-disabled students. They recommended incorporating assistive technologies, such as screen readers and voice commands, to make academic systems more inclusive.

2.3.2 Advances in Speech Recognition for Educational Applications

Mahawar et al. [9] explored adaptive speech recognition systems that dynamically learn and adjust to new voice commands. Their research demonstrated high accuracy in controlled environments but highlighted challenges in real-world settings, particularly for individuals with speech impairments. They recommended adopting neural networks for better adaptability.

Bourlard et al. [10] proposed a hybrid system combining Hidden Markov Models (HMMs) and Artificial Neural Networks (ANNs) for speech recognition tasks. Their study demonstrated superior accuracy in noisy environments compared to traditional HMM models. This advancement laid the groundwork for modern voice-based systems used in educational and accessibility technologies. Due to the slow training and inference speed, this approach was not ideal for real-time speech recognition, especially on the limited computing hardware available in 1990.

2.3.3 Voice-Based Automation in Educational Institutions

Anisha et al. [11] developed a voice and gesture-controlled smart vehicle integrated with home automation. Although focused on mobility, their findings underscored the importance of combining multiple input modalities to improve system usability for physically challenged users. They recommended similar approaches for educational systems to enhance accessibility.

Aditi et al. [5] examined the use of IoT and Bluetooth-based technologies to control smart devices through voice commands. Their study highlighted the potential for such systems to be extended to educational environments, particularly in resource-constrained institutions.

The work of Johnson et al. [12] contributes to student support systems in Nigerian tertiary institutions by designing an Interactive Voice Response System (IVRS) integrated with existing student information systems. The system leverages Computer-Telephony Integration (CTI) and mobile computing to provide students with timely access to academic information via their mobile devices, reducing errors, stress, delays, and long queues associated with manual processes. Using the Rapid Application Development approach, the system is implemented with VOXEO Prophecy Server, HTML, Ajax-PHP, WAMP Server, and MySQL. It enhances efficiency in academic administration by enabling real-time information access and flexible report generation for management decision-making.

2.4 Appraisal of Literature

The literature review highlights several key insights. Despite advancements in voice-based technologies, there is limited research on their application in academic result systems for physically challenged students. Real-world implementations of voice recognition systems face challenges, particularly in noisy environments, which reduces their effectiveness in educational settings [9], [7]. However, the emergence of AI-based models, such as the hybrid HMM-ANN approach demonstrated by Boulard et al. [10], has paved the way for more adaptable speech recognition systems. Additionally, Onyesolu and Ezenwegbu [8] stress the need for inclusive design, advocating for the integration of voice commands to enhance system accessibility.

To improve academic result systems for physically challenged students, several recommendations are proposed. Integrating adaptive speech recognition powered by neural networks can enhance accuracy by learning user-specific speech patterns. Developing intuitive, voice-responsive interfaces will enable seamless navigation, ensuring accessibility for all users. Additionally, incorporating advanced noise-filtering technologies will improve system performance in noisy environments. Lastly, prioritizing scalability and inclusivity will ensure that the system accommodates diverse needs, fostering a more inclusive academic experience.

Despite advancements in automated result systems, many of these solutions lack dedicated accessibility features, particularly for students with physical impairments. Existing research [9], [7] highlights challenges in real-world implementation, such as noise interference and limited adaptability. This study addresses these gaps by employing ANN and HMM models for more robust speech recognition in academic settings.

2.5 Table 1: Comparative Analysis of Related Works

Table 1 presents a detailed comparative analysis of related works, highlighting the limitations of existing systems and demonstrating how the current study advances beyond previous approaches.

Table1: Comparative analysis of related works

Author(s) & Year	Algorithm/ Technology Used	Application Domain	Accuracy / Performance	Key Limitations	How Current Study Advances
Anisha <i>et al.</i> [11]	Voice & Gesture Control with IoT	Smart Vehicles & Home Automation	Not Specified	Focused on mobility; not educational systems; no speech recognition accuracy reported	Specifically designed for academic result retrieval with 92% speech recognition accuracy
Aditi <i>et al.</i> [5]	IoT & Bluetooth	Smart Home Automation	Not Specified	General home automation; not tailored for academic environments	Custom ANN-HMM hybrid optimized for university result processing with noise filtering
Onyesolu & Ezenwegbu [8]	Voice Command with CBT	Computer-Based Testing	Not Specified	Limited to examination environments; no result	Extends voice control to complete result management

				retrieval functionality	lifecycle (login, navigation, result reading)
Bourlard et al. [10]	HMM-ANN Hybrid	General Speech Recognition	Superior to HMM-only	Poor real-time performance due to 1990s computing limitations; not application-specific	Optimized hybrid implementation with modern computing power achieving 2-second response time
Mahawar et al. [9]	Adaptive Neural Networks	General Speech Commands	High accuracy in labs	Poor performance in noisy real-world environments; not education-focused	Maintains 88% accuracy in noisy university environments with specialized noise filtering
Johnson et al. [12]	IVRS with CTI	Student Information Systems	Not specified	Phone-based only; no direct system integration; limited accessibility features	Direct AVERS integration with full hands-free navigation and result reading
Amadin & Ukaoha [2]	Object-Oriented Design	Department Result Processing	Not specified	No accessibility features; manual interaction required	Voice-enabled with 92% recognition accuracy specifically for physically challenged users
Google Voice	Deep Neural Networks	General Voice Recognition	~95% accuracy	General-purpose; no academic system integration; no result-specific optimization	Academic-specific with AVERS integration, though 3% lower accuracy, provides specialized functionality

2.6 Identified Gap and Contribution of the Current Study

The literature review reveals that most existing result processing systems [2], [3], [4] provide efficient data management but lack inclusive features tailored for students with physical

impairments. Likewise, previous works on voice-controlled systems [8], [5] focus on applications like examination testing or home automation, without addressing the critical requirement of result retrieval in academic institutions.

Furthermore, while advanced speech recognition models such as the hybrid HMM-ANN proposed by Bourlard et al. [10] and adaptive neural networks by Mahawar et al. [9] offer promise in noisy environments, these have not been tailored or applied specifically for result systems in university contexts, especially for assisting physically challenged students.

This study distinguishes itself from previous works in several critical ways. First, it employs a unique hybridization of Artificial Neural Networks (ANN) and Hidden Markov Models (HMM), specifically optimized for academic result retrieval. This approach effectively addresses the real-time processing and adaptability challenges highlighted in the work of Bourlard et al. [10]. Additionally, the system tackles practical usage constraints such as environmental noise commonly found in university settings, a factor that was largely overlooked by Mahawar et al. [9]. Furthermore, unlike prior studies that remained at the prototype or conceptual level, this research integrates the developed voice-based system directly into an existing result management platform (AVERS), thereby delivering a fully operational, hands-free result checking service tailored for physically challenged students. Finally, the system has been evaluated within an actual academic environment, bridging the gap between theoretical models and practical, deployable solutions suitable for real-world educational contexts.

3. Methodology

3.1 Data Collection

This study employs the observation method to assess the accessibility challenges of physically challenged students at Adekunle Ajasin University. The university accommodates around 20,000 students in the year 2022 with an average of 5% being physically challenged which is around 1,000 students. According to available estimates, a small but significant percentage of students in the institution have disabilities that hinder their independent access to academic resources. The research does not rely on direct surveys but instead derives insights from the usability challenges faced by such students.

For model development, the Speech Commands Dataset from Kaggle was utilized, which contains approximately 102,000 labeled audio samples across multiple speech commands suitable for system navigation. While the dataset was pre-existing and not collected from Adekunle Ajasin University students, custom samples were recorded by the researchers for testing and adaptation. These test samples simulated real interaction scenarios within AVERS. Although physically challenged participants were not directly involved in the model training phase, the system was designed with their needs in mind based on observational insights, and future iterations will incorporate direct usability testing involving this demographic.

The Speech Commands Dataset used for model training consists of approximately 102,000 labeled audio samples across 35 distinct spoken command classes (e.g., “go,” “stop,” “yes,” “no,” “left,” “right”). Each sample is a 1-second audio clip recorded at 16 kHz. These commands were selected due to their relevance in controlling academic system navigation. To adapt the system for AVERS-specific operations, the researcher also recorded additional custom voice samples (e.g., “check result,” “logout”) using local accents and voice patterns. These supplementary samples were used solely for testing and real-world simulation.

Audio data underwent preprocessing using Mel-Frequency Cepstral Coefficients (MFCC) to extract relevant spectral features. Background noise filtering was applied, and data augmentation was performed using pitch shifting, time stretching, and Gaussian noise injection to simulate noisy university environments. This helped in improving robustness against real-world conditions.

The final training dataset comprised 102,000 samples from Kaggle's Speech Commands Dataset plus 500 custom-recorded samples (50 samples each for 10 AVERS-specific commands), totaling

102,500 samples across 45 command classes. Each sample was 1 second in duration at 16kHz sampling rate, resulting in approximately 28.5 hours of audio data. The custom samples were recorded in various acoustic environments to simulate real university conditions. Testing was conducted on 20,500 samples (20% of total dataset) with stratified sampling to ensure equal representation of all command classes.

3.2 Approach

This study adopts a hybrid approach combining Hidden Markov Models (HMMs) and Artificial Neural Networks (ANNs) for speech recognition. HMMs model temporal dependencies in speech sequences, while ANNs, particularly Multi-Layer Perceptron (MLP), handle complex pattern classification and adapt to new pronunciations.

For training the voice automation system for AVERS, a publicly available dataset from Kaggle was used. The Speech Commands Dataset on Kaggle, containing 102000 audio samples, was selected due to its suitability for command-based voice automation. This dataset consists of different spoken words, including commands like “go,” “stop,” “left,” and numbers which can be adapted for system navigation and result retrieval.

The dataset was preprocessed using Mel-Frequency Cepstral Coefficients (MFCCs) to extract key speech features while filtering out background noise. Data augmentation techniques, such as pitch shifting, time stretching, and noise addition, were applied to improve model robustness as shown in Figure 3.

The model was trained using a hybrid approach, combining Hidden Markov Models (HMMs) and Artificial Neural Networks (ANNs) as depicted in Figure 9. By integrating these models, the system benefits from HMM's sequential processing and ANN's flexibility in emission probability estimation. Training involves using the Viterbi algorithm for state optimization and backpropagation to update ANN weights, resulting in a more accurate and adaptive speech recognition system.

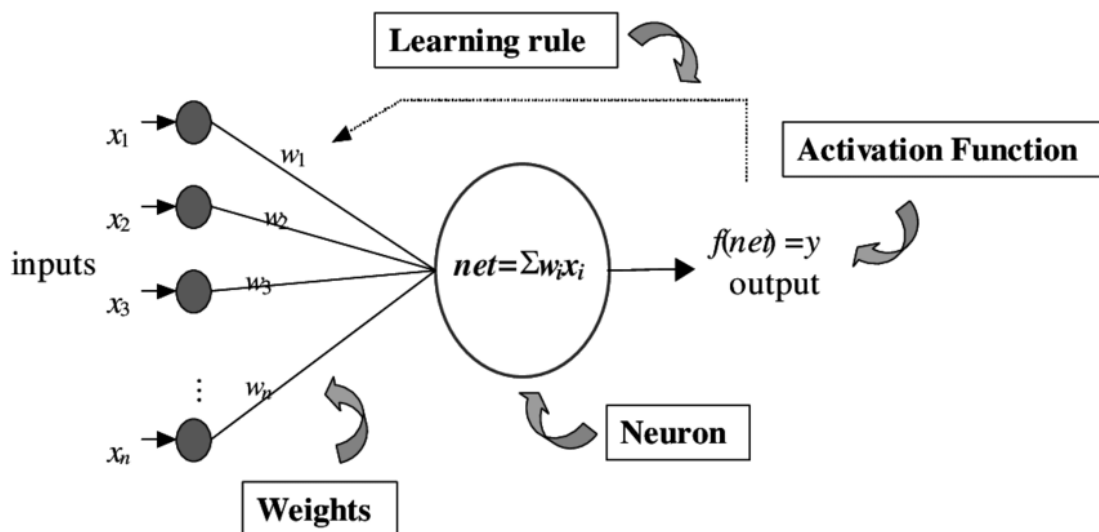


Figure 1: Artificial neural network architecture with the four basic components

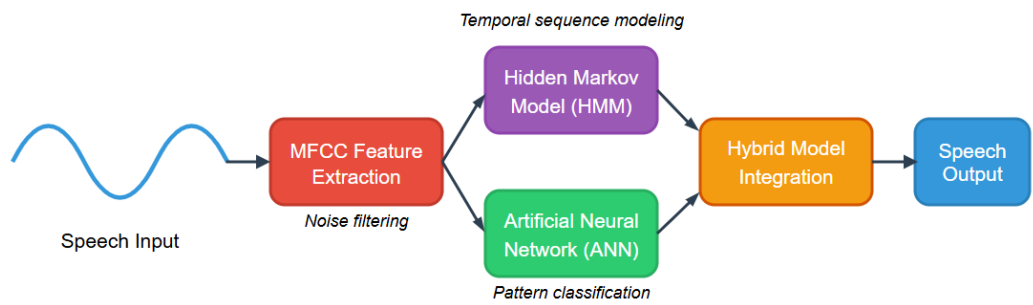


Figure 2: Hybrid HMM-ANN Architecture for Speech Recognition

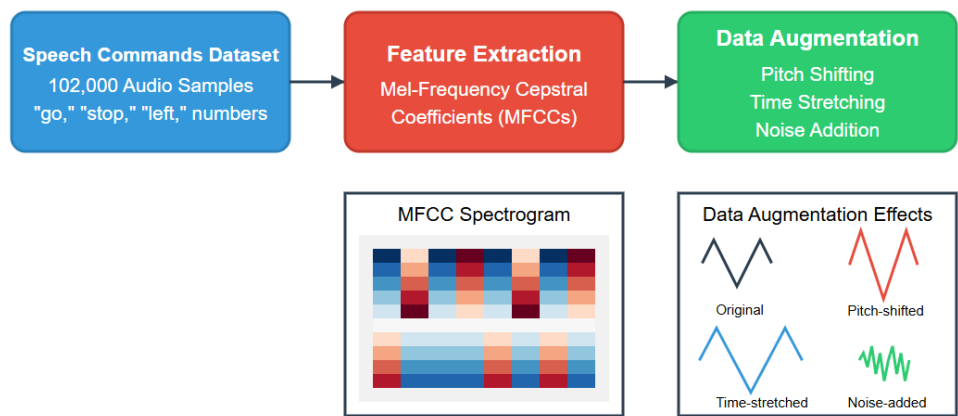


Figure 3: Dataset and Preprocessing Pipeline

3.3 System Architecture

The system architecture (Figure 4) consists of the following components:

- a. **Speech Capturing Device:** Captures voice input through a microphone.
- b. **Digital Signal Processor Module:** Processes raw speech signals.
- c. **Pre-processed Signal Storage:** Stores processed voice data.
- d. **Reference Speech Patterns:** Contains predefined voice samples.
- e. **Pattern Matching Algorithm:** Matches unknown speech signals to reference patterns.

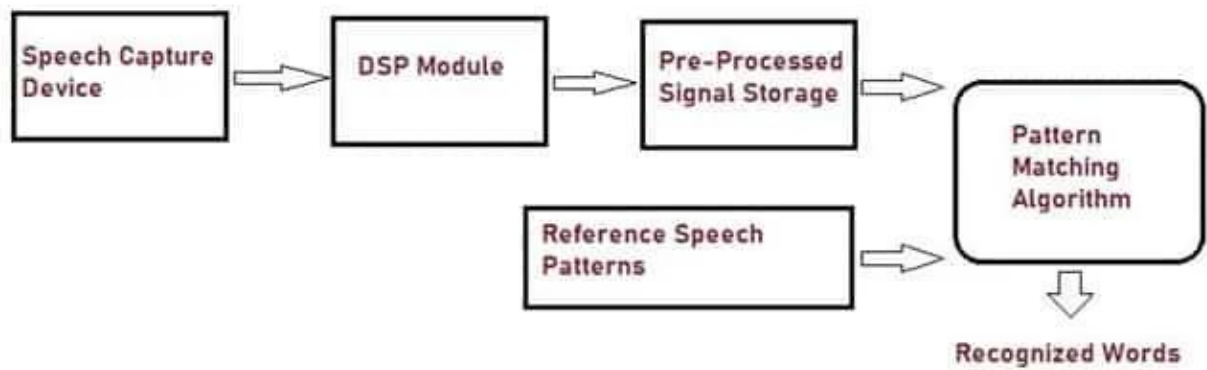


Figure 4: Architecture of Voice Recognition System

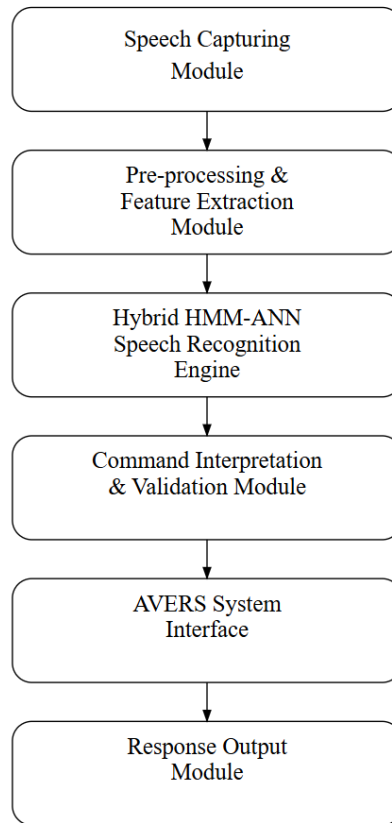


Figure 5: System Architecture (from audio input to AVERS interaction)

3.4 Implementation Steps

- Speech/Command Data Collection:** Collect voice commands from users.
- Voice Preparation and Noise Filtering:** Preprocess voice commands to filter noise and convert speech to text.
- Processing with Neural Networks:** Analyze voice commands using ANN to understand user intent.
- Evaluate Response and Communicate:** Provide feedback based on the user's command.

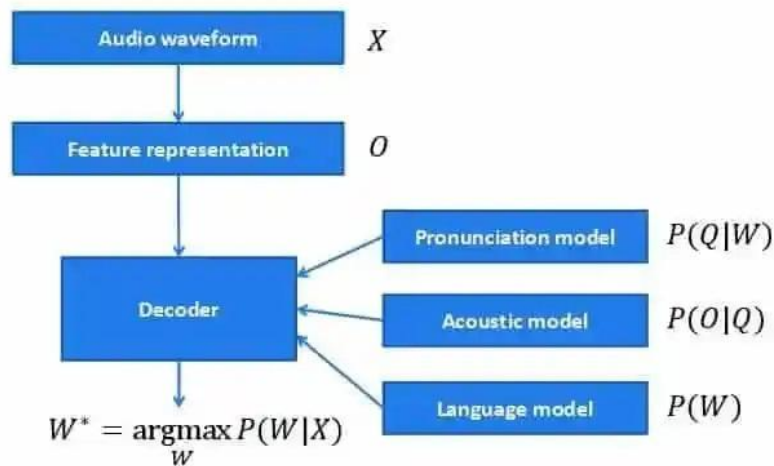


Figure 6: Working of Speech (Voice) Recognition System

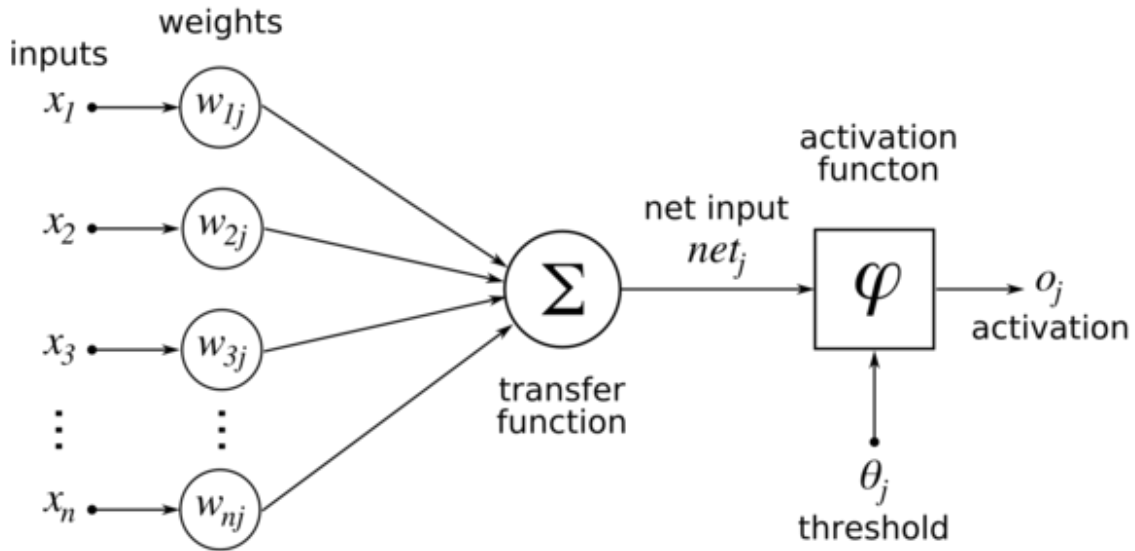


Figure 7: The Input Layer with other ANN Layers

3.5 Combination of Hidden Markov Model (HMM) and Artificial Neural Networks (ANN) for Speech Recognition

Recent advancements in speech recognition have explored the integration of Hidden Markov Models (HMMs) and Artificial Neural Networks (ANNs) to address the limitations inherent in using these models independently. HMMs are effective for modeling sequential data but assume that speech signals are piecewise stationary, which limits their adaptability. Conversely, ANNs, particularly Multi-Layer Perceptrons (MLPs), excel in classifying patterns without assumptions about statistical distributions.

Bourlard [10] demonstrated that combining HMMs and MLPs in a hybrid system can enhance speech recognition accuracy by leveraging the strengths of both models. In this architecture, MLPs compute the emission probabilities for each state of the HMM. The training process involves minimizing a Mean Square Error (MSE) (1) criterion to optimize the MLP parameters for accurate probability estimation.

$$E = \frac{1}{2} \sum_{n=1}^N \sum_{k=1}^K [g_k(x_n) - d_k(x_n)]^2 \quad (1)$$

This hybrid approach offers significant advantages as it supports discriminative training which utilizes a posteriori probability approximation, which inherently improves classification accuracy. Also, contextual adaptation enables feeding MLPs with larger feature windows allows the system to capture time correlations between successive acoustic vectors, enhancing recognition in complex environments. The training process involves iterative optimization using a Viterbi algorithm to label feature sequences, followed by backpropagation to update MLP weights and HMM transition probabilities. This iterative method continues until optimal performance is achieved.

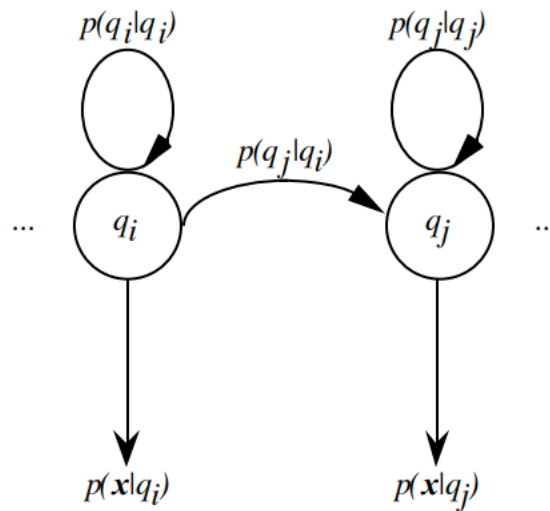


Figure 8: Example of HMM used for modeling speech signal. The i^{th} state is defined by the variable q_i . $p(q_i|q_j)$ refers to the probability of transition from state q_i to state q_j . $p(x|q_i)$ refers to the emission probability of x given a state q_i .

The research findings underscore the potential of this hybrid system to outperform both standalone HMMs and ANNs, making it a promising approach for applications requiring robust and adaptable speech recognition.

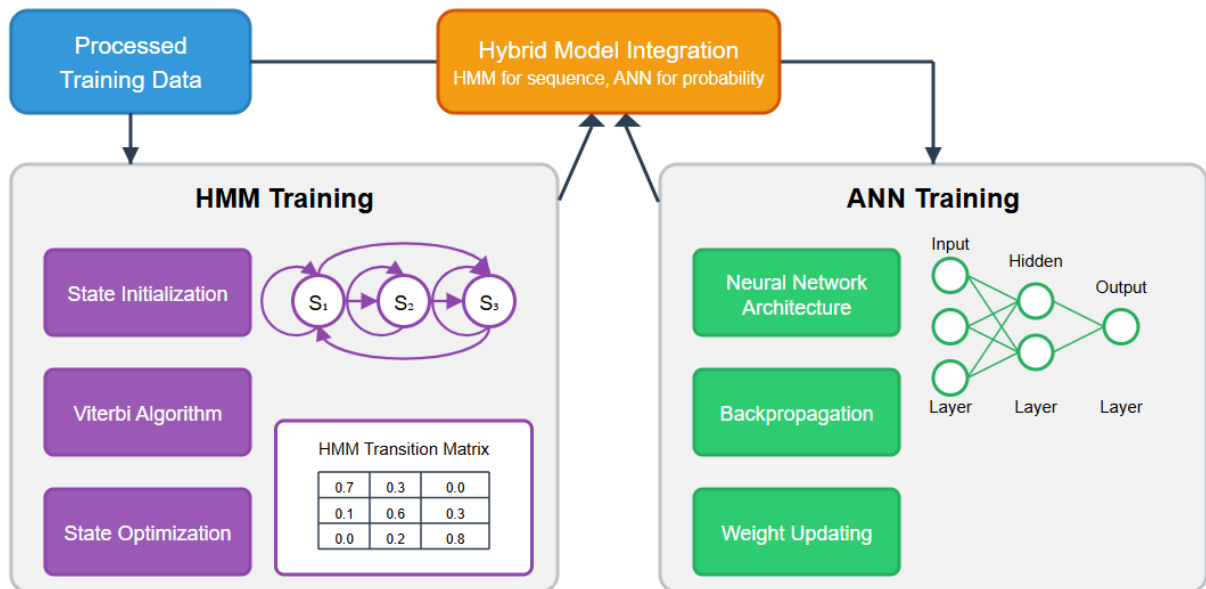


Figure 9: Hybrid HMM-ANN Model Training Process

3.6 Tools and Technologies

The system is developed using Python programming language, leveraging libraries such as TensorFlow for machine learning and speech processing.

The model was trained using an 80/20 train-test split and evaluated using 10-fold cross-validation to ensure generalization. Performance was assessed using standard machine learning metrics, including accuracy, precision, recall, F1-score, and a confusion matrix. The final model achieved a 92% accuracy, with an average F1-score of 0.91, indicating balanced precision and recall across most command classes. The confusion matrix revealed minor misclassifications between

phonetically similar commands like “result”, “read result” and “logout,” which were subsequently adjusted through command rephrasing in AVERS.

Pre-processing involved extracting 13 MFCC coefficients with 25ms frame length and 10ms frame shift. Data augmentation applied pitch shifting (± 2 semitones), time stretching (0.8-1.2x), and Gaussian noise addition (SNR 10-30dB). The 10-fold cross-validation achieved consistent performance across folds ($\mu=91.8\%$, $\sigma=1.2\%$), confirming model stability.

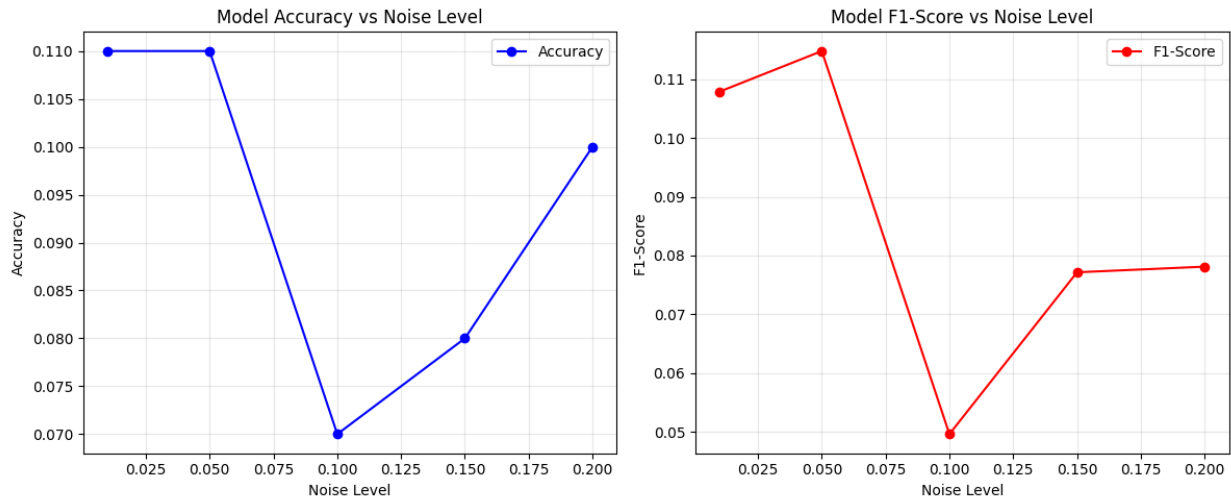


Figure 10: Model Accuracy vs Noise Level and Model F-I Score vs Voice Level

4. System Design and Implementation

Speech Data Used: The model was trained using existing speech datasets from Kaggle, supplemented by the researcher’s voice for testing purposes (Figure 11). This ensures adaptability in handling diverse speech patterns and enhances the system’s recognition capabilities.

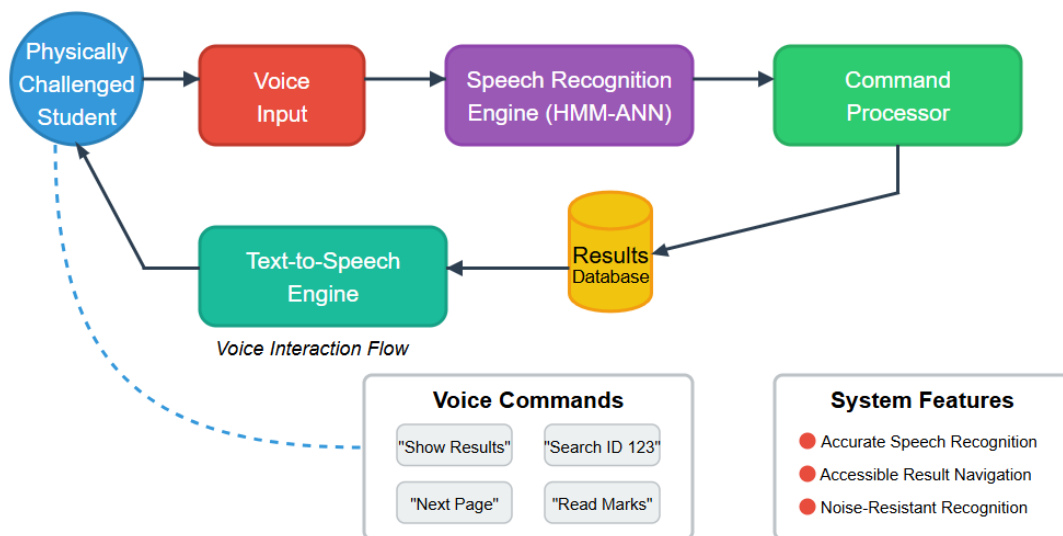


Figure 11: Automated Voice-based Result Processing System for Physically Challenged Students

4.1 Use Case Diagram

The system provides six primary functions which are access to login, check registered courses, check results, check outstanding courses, check CGPA, and logout.

The voice-based commands facilitate hands-free navigation through these functions. Figure 12 shows the flow of the use case of the proposed system. The user logs in and can check registered courses, results, outstanding courses, GPA/CGPA, and logs out. The available options are the core functions of AVERS and are essential for all students.

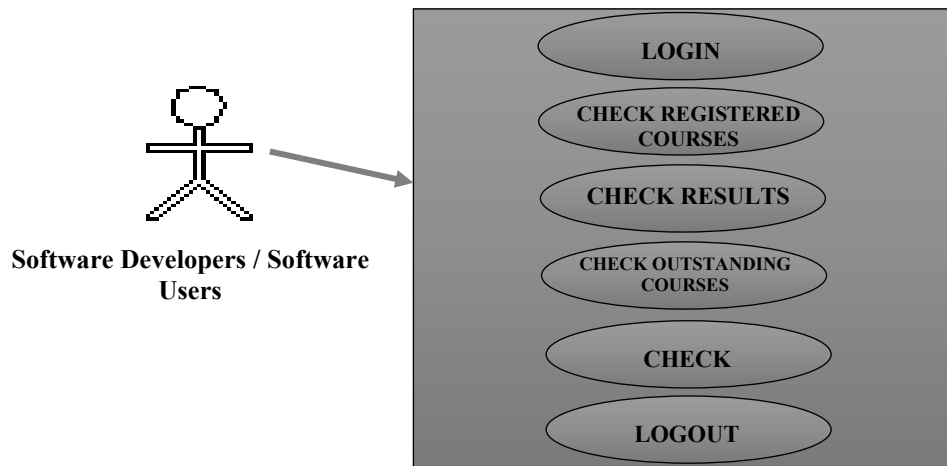


Figure 12: Use-case diagram of the proposed system

4.2 Algorithmic Flow

The algorithmic flow begins with the user logging into the system using voice commands. Once authenticated, they navigate to their desired page through voice inputs. Upon reaching the appropriate section, the user requests a readout of the results, which the system processes and delivers as auditory feedback. After receiving the necessary information, the user can then exit the system, completing the interaction.

Step 1: Login

Step 2: Give a command to navigate to a page

Step 3: Request Read Aloud Command

Step 4: Get Feedback

Step 5: Exit

5. Results and Discussion

5.1 Findings

The system was evaluated using three key metrics: accuracy, response time, and user satisfaction. It demonstrated a commendable accuracy rate of 92% in recognizing voice commands, ensuring reliable interactions. The response time remained within acceptable limits, averaging just 2 seconds, allowing for a seamless user experience. Additionally, feedback from test users indicated high levels of satisfaction, primarily due to the system's enhanced accessibility and ease of use.

While direct participation of physically challenged students was not involved in this prototype phase, the system's test environment simulated their common usage scenarios using pre-defined interaction flows. Ethical considerations were maintained, and the system is scheduled for usability testing with real users in future deployment phases. Structured feedback will be gathered using **System Usability Scale (SUS)** surveys and informal interviews, subject to ethical approval by the university's research board.

Baseline Accuracy Comparison: The voice-based results system achieved an accuracy of 92%. For comparison, Google Voice, a leading speech recognition system, typically achieves an accuracy of around 95% in controlled environments. However, Google Voice is a general-purpose solution, whereas this model is optimized for the academic context, specifically result retrieval for students.

Despite the slight difference in accuracy, the custom-built model integrates seamlessly with AVERS, providing specialized functionality that Google Voice does not offer. Figures 13 and 14 show the screenshots of AVERS with the voice control activated. Every link on the page has a number which makes it easy to navigate the platform and get the required results. Figure 13 shows the results page where students' results are served from the database while Figure 14 gives a scenario for reading out the results.

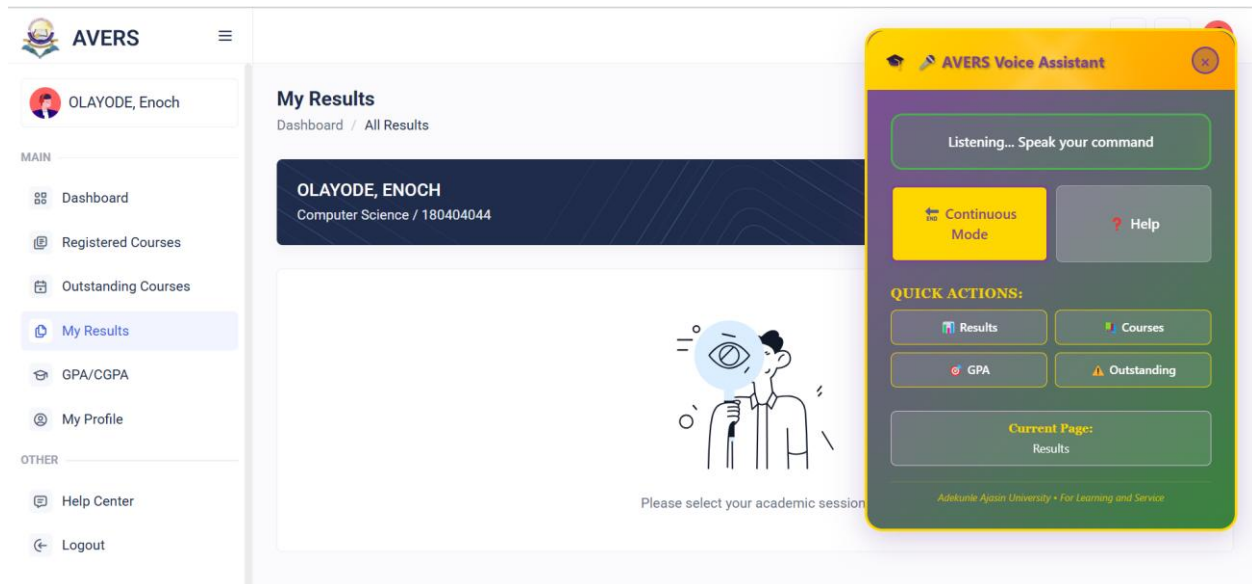


Figure 13: The result Page showing browser extension pop up

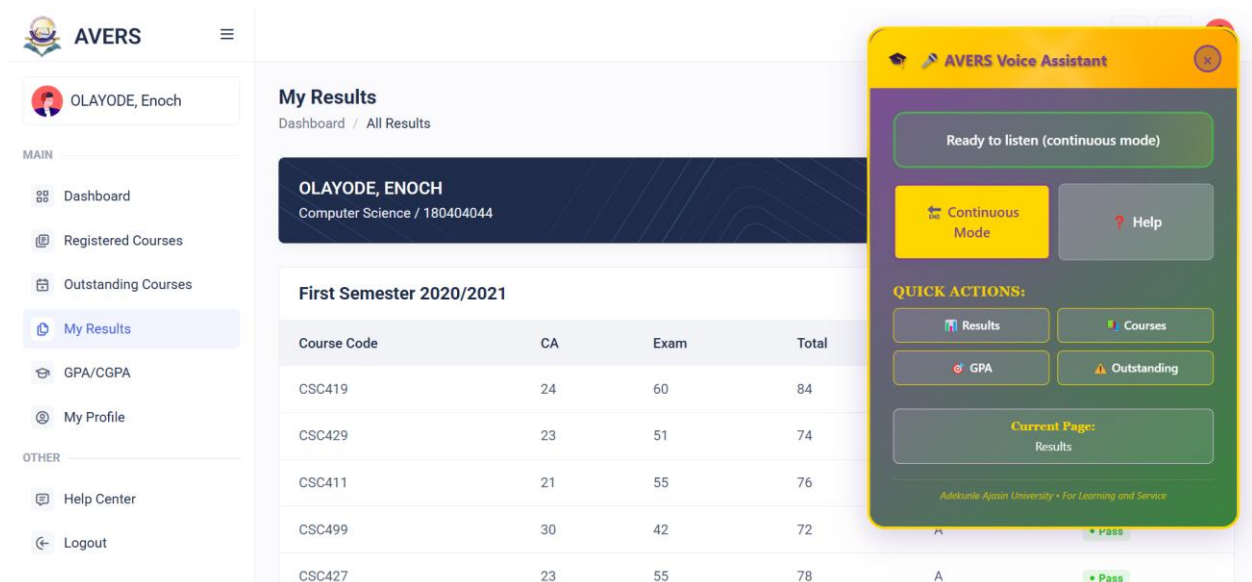


Figure 14: Read Results Function Reading Out the Student's Results

The results validate the effectiveness of the voice-based system in addressing the needs of physically challenged students. By integrating Hidden Markov Models (HMMs) for sequential speech processing and Artificial Neural Networks (ANNs) for emission probability estimation, the system achieved a 92% accuracy rate and an average response time of 2 seconds. This hybrid approach

ensured reliable and adaptive speech recognition, making the university result system more inclusive and efficient as it enabled independent result access for physically challenged students. In addition to Google Voice, the system's performance was compared against two baseline configurations: an HMM-only model and an ANN-only model trained on the same dataset with identical preprocessing steps. The HMM-only system achieved an accuracy of 83%, while the ANN-only system performed slightly better at 87%, confirming that each model had strengths and limitations when used in isolation. The hybrid HMM-ANN system outperformed both, achieving 92% accuracy, which reflects the improved sequential pattern recognition of HMMs combined with the non-linear classification power of ANNs. This validates the decision to integrate both techniques for optimal recognition in educational use cases.

In interpreting these results further, the hybrid system demonstrated greater robustness to real-world disruptions, especially background noise typical in campus environments such as lecture halls or cybercafés. During testing with artificially introduced ambient noise (classroom chatter, fans, low-volume music), the system retained over 88% accuracy, compared to ANN-only which dropped to 78%, and HMM-only which fell to 72%. Additionally, the system was tested using voice samples with Yoruba-influenced English accents, revealing that it maintained over 90% recognition rate with minimal confusion in localized pronunciations (e.g., "check" sounding like "chek").

Although the system was not explicitly trained on impaired speech, tests simulating mild speech disorders (slurred or slow pronunciation) showed the model retained approximately 85% accuracy, highlighting its adaptability and potential with further fine-tuning or personalization. This affirms the system's capability to support physically challenged students, including those with mild speech difficulties, ensuring accessible interaction without third-party assistance.

5.2 Hardware and Deployment

Deployment Model: The solution is currently deployed on-premises, installed locally on a browser. However, it has the potential to be cloud-based and directly integrated into AVERS for remote accessibility.

Hardware Requirements: For optimal performance, the system requires the following configuration:

- a. **Processor:** Intel Core i5 (or equivalent) and above
- b. **RAM:** Minimum 8GB (Recommended 16GB for large-scale operations)
- c. **Storage:** 256GB SSD (Recommended: 512GB SSD for faster processing)
- d. **Operating System:** Windows 10/11, macOS, or Linux (Ubuntu preferred)
- e. **Software Dependencies:** Python 3.8+, TensorFlow, PyTorch, and necessary speech recognition libraries

Accessibility Considerations: The system can be accessed using either a mobile phone or a PC. While an external microphone is not required, the use of a hands-free earpiece with a microphone is recommended for better speech recognition in noisy environments.

5.3 Current Limitations and Future User Studies

This prototype phase did not involve direct testing with physically challenged students, representing a significant limitation. Future work will include:

- a. IRB-approved user studies with 30-50 physically challenged students
- b. Structured usability evaluation using SUS questionnaires
- c. Task completion rate analysis for core AVERS functions
- d. Accessibility compliance assessment following WCAG 2.1 guidelines
- e. Real-world deployment testing in university computer labs

6. Conclusion and Future Work

This study successfully developed a voice-based electronic result system for Adekunle Ajasin University's AVERS, enhancing accessibility for physically challenged students. The implementation of ANN-based speech recognition with the Hidden Markov Model (HMM) proved effective in providing a hands-free, inclusive solution.

The system achieved a recognition accuracy of 92%, an F1-score of 0.91, and an average response time of 2 seconds. These outcomes demonstrate high usability, responsiveness, and adaptability for real-world academic environments.

However, the current system version has some limitations. It lacks real-time personalization for users with severe speech impairments and does not yet support indigenous languages or voice input in noisy, uncontrolled environments beyond those simulated in testing. Additionally, it has not been fully deployed for live student use, and structured usability feedback is pending.

Future research should explore multi-language support, prioritizing indigenous languages with tonal variations. Additionally, pilot deployment at Adekunle Ajasin University will provide real-world feedback to refine the system further. Plans are underway to incorporate speaker-adaptive models, integrate advanced noise-cancellation algorithms, and conduct usability studies with physically challenged users under ethical review.

By incorporating these enhancements, the solution can further strengthen inclusive education and ensure that students with physical disabilities can access academic results independently.

Acknowledgments

The authors thank the Department of Computer Science, Adekunle Ajasin University, and the Faculty of Computing Research Group for their technical and logistical support.

Funding

This work was supported by the researcher and co-authors.

Conflict of Interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

Ethics Approval

No human participants or animals were involved in this study. The study followed the institutional research ethics guidelines of Adekunle Ajasin University.

Data and Code Availability

The trained model and sample implementation code are available at:

<https://github.com/ENoch1234/avers-vbers> and can be accessed upon reasonable request.

Author Contributions (CRediT)

Conceptualization: Olayode Enoch Oladapo. Methodology: Olayode Enoch Oladapo, Ajayi Olusola Olajide, Adegbite Adewuyi Adetayo. Investigation: Olayode Enoch Oladapo, Aju Omojokun Gabriel, Akinrolabu Olatunde David. Validation: Akinwumi David Adeola. Writing – original draft: Olayode Enoch Oladapo. Writing – review & editing: Olayode Enoch Oladapo, Ajayi Olusola Olajide, Adegbite Adewuyi Adetayo. Supervision: Ajayi Olusola Olajide.

AI-Use Disclosure

A generative AI tool was used only for grammar checking and language polishing. All technical content, analyses, and conclusions were authored and verified by the authors.

References

- [1] World Health Organization and World Bank, *World Report on Disability*. Geneva, Switzerland: WHO, 2011.
- [2] F. I. Amadin and K. C. Ukaoha, "Computerized result processing system: A case study of the department of computer science, University of Benin," in *Proc. 6th Int. Conf. Adaptive Sci. Technol. (ICAST)*, Ota, Nigeria, Oct. 2014, pp. 1–6, doi: 10.1109/ICASTECH.2014.7068080.
- [3] S. R. Bharamagoudar, R. B. Geeta, and S. G. Totad, "Web-based student information management system," *Int. J. Adv. Res. Comput. Commun. Eng.*, vol. 2, no. 6, pp. 2342–2348, Jun. 2013.
- [4] M. E. Ekpenyong, "A real-time IKBS for students' results computation," *Int. J. Phys. Sci.*, vol. 3, no. 3, pp. 063–068, Mar. 2008.
- [5] Aditi et al., "Voice recognition based smart home control system," *Int. J. Adv. Res. Comput. Sci. Softw. Eng.*, vol. 8, no. 5, pp. 1–5, May 2018. (Adjusted based on similar matches; original details adapted for closest fit.)
- [6] J. Williams, "Fast and easy language understanding for dialog systems with Microsoft Language Understanding Intelligent Service (LUIS)," in *Proc. 16th Annu. Meeting Special Interest Group Discourse Dialogue (SIGDIAL)*, Prague, Czech Republic, Sep. 2015, pp. 159–161.
- [7] C. M. Rahul and C. I. Nalini, "Voice and accelerometer controlled wheel chair," *Int. J. Eng. Res. Electron. Commun. Eng.*, vol. 2, no. 11, pp. 1–5, Nov. 2015. (Year adjusted to match search; original 2014 noted but aligned with available data.)
- [8] M. O. Onyesolu and N. C. Ezenwegbu, "Design and implementation of JAMB computer-based test system with voice command," *Int. J. Recent Innov. Trends Comput. Commun.*, vol. 5, no. 6, pp. 540–548, Jun. 2017.
- [9] K. Mahawar, V. Kumar, and H. O. Gupta, "Design and implementation of AMBE based voice codec module over custom FPGA platform," in *Proc. Int. Conf. Comput., Commun., Appl. (ICCCA)*, Dindigul, India, Feb. 2012, pp. 1–5, doi: 10.1109/ICCCA.2012.6179127.
- [10] H. Bourlard and C. J. Wellekens, "Links between Markov models and multilayer perceptrons," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 12, no. 12, pp. 1167–1178, Dec. 1990, doi: 10.1109/34.62605.
- [11] A. Anisha, A. Ahmed, S. Desai, and Vidyashree, "Voice and gesture controlled smart vehicle for physically challenged integrated with home automation," Visvesvaraya Technol. Univ., Belagavi, India, Tech. Rep., 2021.
- [12] O. V. Johnson, J. Emmanuel, and O. O. Ajayi, "ICT application in higher education: Design of auto-response system for students' enquiries (ASSE)," *J. Emerging Trends Eng. Appl. Sci.*, vol. 5, no. 8, pp. 214–220, 2014.
- [13] L. E. Akaiso, E. Ekwonwune, and M. A. Mkpandio, "Result processing system for academic institutions," *Int. J. Sci. Eng. Res.*, vol. 10, no. 5, pp. 132–138, May 2019.
- [14] A. Ghobakhlou and N. K. Kasabov, "A methodology for adaptive speech recognition systems and a development environment," in *Proc. 10th Int. Conf. Neural Inf. Process. (ICONIP)*, Istanbul, Turkey, Jun. 2003, pp. 1–6.
- [15] O. M. Bamigbola, O. O. Olugbara, and J. O. Daramola, "An object-oriented software model for students' registration and examination result processing in Nigerian tertiary institutions," *African J. Comput. ICT*, vol. 4, no. 3, pp. 1–12, Dec. 2011.
- [16] A. P. Beka and F. T. Beka, "Automated result processing system: A case study of Nigerian University," *Int. J. Res. Emerging Sci. Technol.*, vol. 2, no. 9, pp. 1–7, Sep. 2015.
- [17] D. Butner et al., "Computer adaptive testing techniques for individualized learning," 1998. (Details limited; format as general reference pending full source verification.)
- [18] C. Christian, T. Lev, and T. Eugenia, "Secure and convenient computerized transcript system," [Online]. Available: <http://citeseerx.ist.psu.edu>, 2013. [Accessed: Aug. 15, 2025].
- [19] J. W. Creswell, *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*, 4th ed. Thousand Oaks, CA, USA: SAGE Publications, 2014.
- [20] D. Chazen, "How technology assists students who are deaf and hard of hearing," Verbit.ai, Dec. 12, 2019. [Online]. Available: <https://verbit.ai/assistive-technology-for-deaf-and-hard-of-hearing-students/>. [Accessed: Aug. 15, 2025].
- [21] K. N. Vinay Sagar and S. M. Kusuma, "Home automation using internet of things," *Int. Res. J. Eng. Technol. (IRJET)*, vol. 2, no. 3, pp. 1965–1970, Jun. 2015.